

Computational Fluid Dynamics for Aerospace Applications

Lecture 9

Compressible Flows: Part I

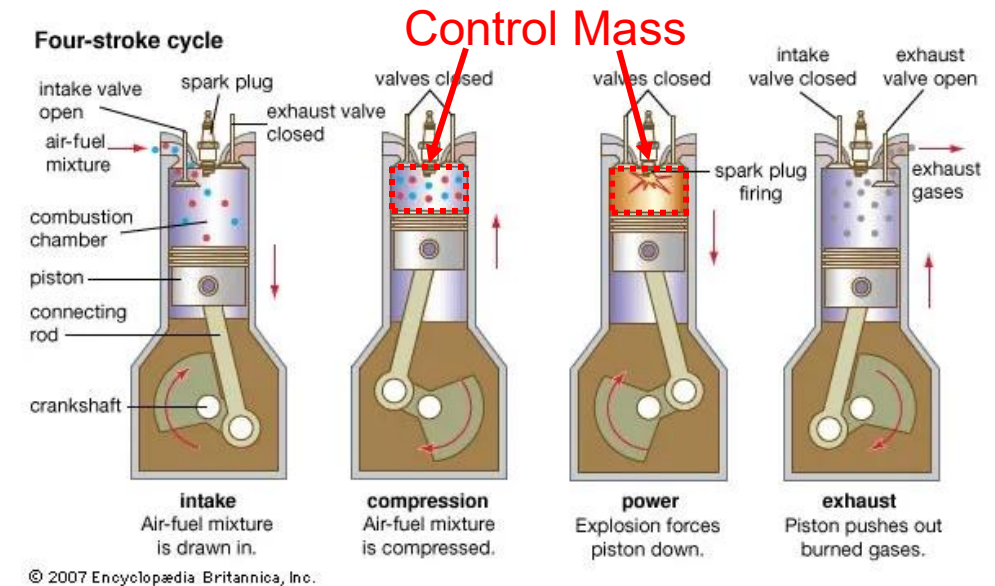
Compressible Flows: Part I

- Fluid Conservation Principles
- Speed of Sound
- Isentropic Flow
- Stagnation Properties

Fluid Conservation Principles

- Control Mass (Closed system)

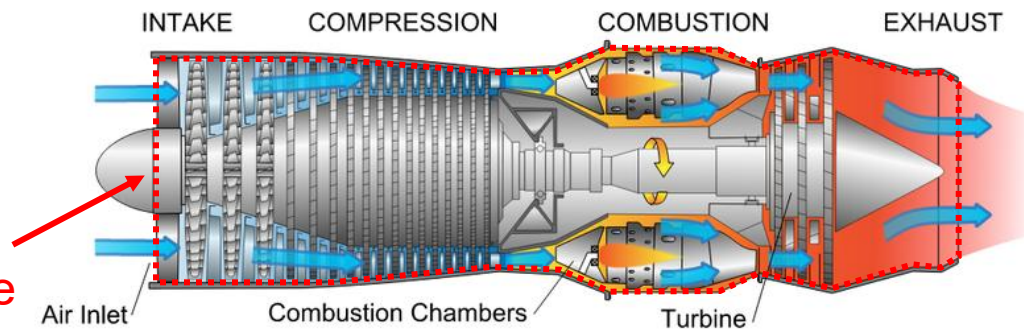
- Mass within system is fixed
 - Mass does not cross boundaries
- Example: Gasoline Engine



- Control Volume (Open system)

- Volume within system is fixed
 - Mass freely crosses boundaries
- Example: Gas turbine

Control Volume



Fluid Conservation Principles

- Reynold's transport theorem

$$\underbrace{\frac{DN}{Dt}}_{\text{Total rate of change of } N} = \underbrace{\frac{\partial}{\partial t} \iiint_{CV} \eta \rho dV}_{\text{Change of } N \text{ in control volume}} + \underbrace{\iint_{CS} \eta \rho (\vec{V} \cdot \hat{n}) dA}_{\text{Flux of } N \text{ across boundaries}}$$

Total rate of
change of N

Change of N in
control volume

Flux of N across
boundaries

- N = extensive property
- $\eta = N/m$

V is volume

$\vec{V}$ is velocity

for now...

* $\hat{n}$ points out

Fluid Conservation Principles

- Mass

- Generally: $\frac{Dm}{Dt} = 0$

- $N = m$

- $\eta = \frac{m}{m} = 1$

- $0 = \frac{\partial}{\partial t} \iiint_{CV} \rho dV + \iint_{CS} \rho (\vec{V} \cdot \hat{n}) dA$

- $0 = \iint_{CS} \rho (\vec{V} \cdot \hat{n}) dA \quad (\text{steady state})$

Fluid Conservation Principles

- Momentum

- Generally: $\frac{Dm\vec{V}}{Dt} = \vec{F}_{net}$

- $N = m\vec{V}$

- $\eta = \vec{V}$

- $\vec{F}_{net} = \frac{\partial}{\partial t} \iiint_{CV} \rho \vec{V} dV + \iint_{CS} \rho \vec{V} (\vec{V} \cdot \hat{n}) dA$

- $\vec{F}_{net} = \iint_{CS} \rho \vec{V} (\vec{V} \cdot \hat{n}) dA$ (steady state)

Fluid Conservation Principles

- Energy

- Generally: $\frac{DE}{Dt} = \dot{Q} - \dot{W} = \dot{Q} - \underbrace{\oint_{CS} p(\vec{V} \cdot \hat{n})dA}_{\text{Pressure work on surface}} - \underbrace{\dot{W}_{\text{other}}}_{\text{Shaft, viscous, etc.}}$

- $N = E$

- $\eta = e = u + \frac{1}{2}\vec{V} \cdot \vec{V}$

- $\dot{Q} - \dot{W}_{\text{other}} = \frac{\partial}{\partial t} \iiint_{CV} \rho \left(\overset{\text{internal energy}}{u} + \overset{e}{\frac{\vec{V} \cdot \vec{V}}{2}} \right) dV + \oint_{CS} \rho \left(u + \frac{\vec{V} \cdot \vec{V}}{2} + \underbrace{\left[\frac{p}{\rho} \right]}_e \right) (\vec{V} \cdot \hat{n}) dA$

- $\dot{Q} - \dot{W}_{\text{other}} = \oint_{CS} \rho \left(u + \frac{\vec{V} \cdot \vec{V}}{2} + \frac{p}{\rho} \right) (\vec{V} \cdot \hat{n}) dA \quad (\text{steady state})$

Fluid Conservation Principles

• Example: Steady pipe flow

• Mass

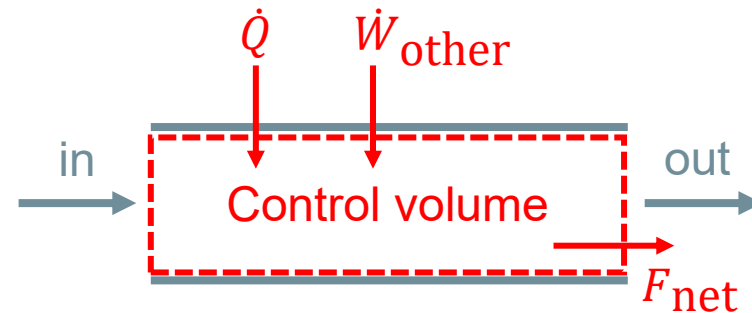
~~*~~ • $\dot{m}_{out} = \dot{m}_{in}$

• Momentum

$$F_{net} = (\dot{m}V)_{out} - (\dot{m}V)_{in}$$

• Energy

$$\dot{Q} + \dot{W}_{other} = \left[\dot{m} \left(u + \frac{V^2}{2} + \frac{p}{\rho} \right) \right]_{out} - \left[\dot{m} \left(u + \frac{V^2}{2} + \frac{p}{\rho} \right) \right]_{in}$$



Mass (Steady)

$$0 = \oint \rho (\vec{V} \cdot \hat{n}) dA$$

$$\vec{V} = 0 \text{ top/bottom}$$

$$(-pVA)_{in} + (pVA)_{out} = 0$$

Momentum (Steady)

$$\oint \rho \vec{V} (\vec{V} \cdot \hat{n}) dA = F_{net}$$

$$-(\rho V^2 A)_{in} + (\rho V^2 A)_{out} = F_{net}$$

$$-(\dot{m}V)_{in} + (\dot{m}V)_{out} = F_{net}$$

Energy (Steady)

$$\dot{Q} - \dot{W} = \oint \rho \left(u + \frac{V^2}{2} + \frac{p}{\rho} \right) (\vec{V} \cdot \hat{n}) dA$$

$$\dot{Q} + \dot{W}_{other} = - \left[\rho \left(u + \frac{V^2}{2} + \frac{p}{\rho} \right) VA \right]_{in}$$

$$+ \left[\rho \left(u + \frac{V^2}{2} + \frac{p}{\rho} \right) VA \right]_{out}$$

Speed of Sound

Isentropic gas relations

Isentropic, ideal gas, constant C_p

$$\left(\frac{p}{p_0}\right) = \left(\frac{\rho}{\rho_0}\right)^\gamma \Rightarrow p = C \rho^\gamma$$

- Sound waves are infinitesimal pressures waves
 - Reversible and adiabatic (therefore also isentropic)

- Isentropic speed of sound a

$$a \equiv \sqrt{\frac{\partial p}{\partial \rho}_s} \quad \leftarrow \text{definition}$$

- For an Ideal Gas (constant specific heats)

$$a = \sqrt{\gamma \frac{p}{\rho}} = \sqrt{\gamma R T}$$

- Mach Number: $M = V/a$

$$\begin{aligned} \frac{dp}{d\rho} &= C \gamma \rho^{\gamma-1} \\ &= C \gamma \frac{p}{\rho} \\ &= \gamma \frac{p}{\rho} \end{aligned}$$

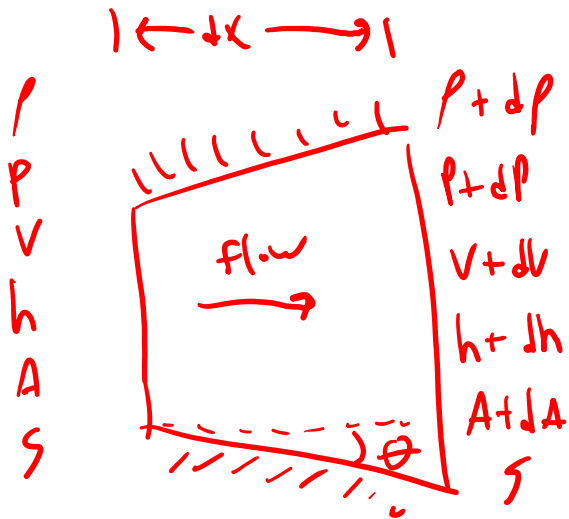
$$a = \sqrt{\frac{\gamma p}{\rho}} + p = \rho R T \rightarrow a = \sqrt{\gamma R T}$$

Isentropic Flow

θ small, and $A = f(x)$

- “One-dimensional” flow, no friction, no heating $\rightarrow$ isentropic

- Mass



Mass (Steady)

$$\oint \rho(\vec{v} \cdot \hat{n}) dA = 0$$

$$(\rho + d\rho)(V + dV)(A + dA) - \rho VA = 0$$

$$\boxed{\rho VA} + \rho V dA + \rho dVA + \underbrace{\rho dV dA}_{\text{H.O.T.}} + d\rho VA + \underbrace{d\rho dV dA}_{\text{H.O.T.}} + \underbrace{d\rho dV dA}_{\text{H.O.T.}} - \boxed{\rho VA} = 0$$

$$\rho V dA + \rho dVA + d\rho VA = 0$$

divide ρVA

$$\frac{dA}{A} + \frac{dV}{V} + \frac{d\rho}{\rho} = 0$$

$\rightarrow$ reverse product rule

$$d(\rho VA) = 0$$

$$d(\dot{m}) = 0$$

$$\bullet \frac{d\rho}{\rho} + \frac{dV}{V} + \frac{dA}{A} = 0$$

* Recall: $\frac{dA}{A} + \frac{dV}{V} + \frac{d\rho}{\rho} = 0$

Isentropic Flow

- “One-dimensional” flow, no friction, no heating
 - Momentum (steady, x-direction)

$$\oint \rho \vec{v} (\vec{v} \cdot \hat{n}) dA = \sum F$$

$$(p + dp)(V + dV)(A + dA) - pV^2 A = \sum F_x$$

Expand, neglect H.O.T.

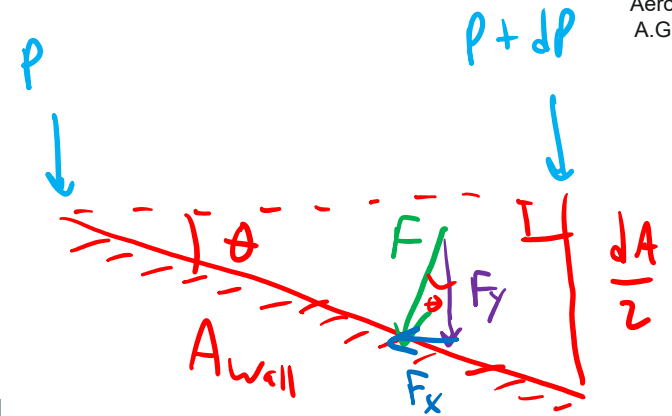
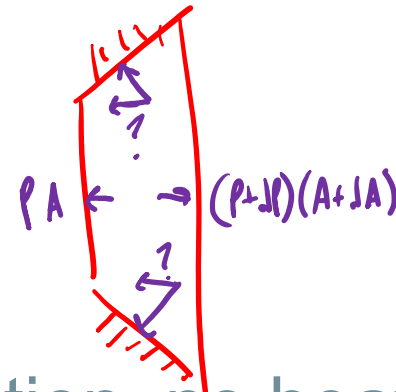
$$\boxed{pV^2 A} + pV^2 dA + 2pVA dV + V^2 A dp - \boxed{pV^2 A} = \sum F_x$$

$$pV^2 A \left(\frac{dA}{A} + \frac{2dV}{V} + \frac{d\rho}{\rho} \right) = \sum F_x$$

$$pV^2 A \left(\frac{dV}{V} \right) = \sum F_x$$

$$\bullet dp + \rho V dV = 0$$

$$pV^2 A \left(\frac{dV}{V} \right) = -A dp \rightarrow pV dV = -dp$$



Assume:

- θ very small
- Pressure varies linearly across wall
- Average pressure on wall is $p + dp/2$

Trig:

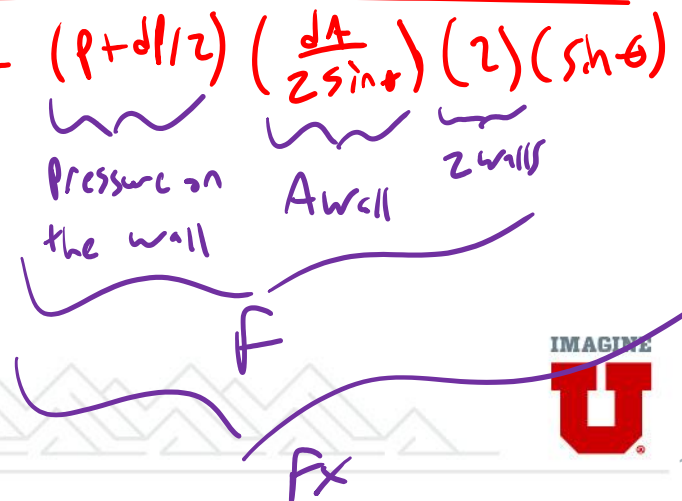
$$\sin \theta = \frac{dA/2}{A_{wall}} \Rightarrow A_{wall} = \frac{dA}{2 \sin \theta}$$

$$\sin \theta = \frac{F_x}{F} \Rightarrow F_x = F \sin \theta$$

$$\sum F_x = pA - (p + dp)(A + dA) + (p + dp/2) \left(\frac{dA}{2 \sin \theta} \right) (2) (\sin \theta)$$

$$\sum F_x = \boxed{pA - pA} - \boxed{Adp} - \boxed{pdA} - \boxed{dpdA} + \boxed{pdA} + \frac{dpdA/2}{\text{H.O.T.}}$$

$$\sum F_x = -Adp$$



Isentropic Flow

$$\text{Continuity: } \frac{dA}{A} + \frac{dV}{V} + \frac{d\rho}{\rho} = 0$$

$$\frac{dA}{A} + \frac{dV}{V} + \frac{d\rho}{\rho} = \frac{d\rho}{\rho} - \frac{d\rho}{\rho}$$

- “One-dimensional” flow, no friction, no heating

- Energy (steady)

$$\oint \rho \left(e + \frac{1}{2} V^2 \right) (\vec{V} \cdot \hat{n}) dA = 0$$

$$\oint \rho e (\vec{V} \cdot \hat{n}) dA + \oint \rho \left(\frac{1}{2} V^2 \right) (\vec{V} \cdot \hat{n}) dA = 0$$

$$(\rho + d\rho)(e + de)(V + dV)(A + dA) - \rho e V A + (\rho + d\rho)(V + dV)(A + dA) - \rho V A = 0$$

Expand, neglect $H.O.T.$, combine, rearrange

$$e(\rho V dA + \rho A dV + V A d\rho) + \rho V A de + (P V dA + P A dV + V A dP) = 0$$

Product rule

$$e d(\rho A) + \rho V A de + V P A \left[\frac{dA}{A} + \frac{dV}{V} + \frac{dP}{P} \right] = 0$$

$\underbrace{e d(\rho A)}_{\dot{m}=0} + \rho V A de + V P A \left[\frac{dA}{A} + \frac{dV}{V} + \frac{dP}{P} \right] = 0$

$\bullet dh + V dV = 0 \Rightarrow d \left(\frac{\gamma}{\gamma-1} RT + \frac{1}{2} V^2 \right) = 0$

Continuity...

$$\rho V A de + V P A \left[\frac{dP}{P} - \frac{dP}{P} \right] = 0$$

$$de = - \frac{P}{\rho} \left[\frac{dP}{P} - \frac{dP}{P} \right] \quad (1)$$

$$e = u + \frac{1}{2} V^2 \quad \left. \begin{array}{l} e = h - P/\rho + \frac{1}{2} V^2 \\ h = u + P/\rho \end{array} \right\} \downarrow \text{differentiate}$$

$$(2) de = dh - dP/\rho + P d\rho/\rho^2 + V dV$$

$$(1) = (2)$$

$$- \frac{P}{\rho} \left[\frac{dP}{P} - \frac{dP}{P} \right] = dh - \frac{dP}{\rho} + \frac{P d\rho}{\rho^2} + V dV$$

$$0 = dh + V dV$$

$$\left. \begin{aligned} \gamma &= c_p / c_v \\ R &= c_p - c_v \end{aligned} \right\} c_p = R \frac{\gamma}{\gamma - 1}$$

Isentropic Flow

- “One-dimensional” flow, isentropic, **no work or heat**

- $d \left(h + \frac{1}{2} V^2 \right) = d \left(\frac{\gamma}{\gamma - 1} R T + \frac{1}{2} V^2 \right) = 0$

- First Law (open system) $\rightarrow c_p$

- $d \left(h + \frac{1}{2} V^2 \right) = \delta q - \delta w$

- Complicated accounting

- If we add heat/work, does it change the kinetic energy or enthalpy?

- We can avoid the distinction by creating a new property that accounts for enthalpy and kinetic energy

Stagnation Properties

- Thermodynamic properties of a gas brought to rest isentropically and adiabatically
- Stagnation Enthalpy

Stagnation enthalpy $h_t = \underbrace{h}_{\text{static enthalpy}} + \frac{1}{2} V^2$

- Stagnation Temperature

$$T_t = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$$

ideal gas

$$a = \sqrt{\gamma R T}$$

$$a^2 = \gamma R T$$

$$M = \frac{V}{a}$$

$$h_t = h + \frac{1}{2} V^2$$

$$h_t - h = \frac{1}{2} V^2$$

Assume C_p
is constant

$$C_p (T_t - T) = \frac{1}{2} V^2$$

$$T_t = T + \frac{V^2}{2 C_p}$$

$$T_t = T + \frac{V^2}{2} \frac{\gamma-1}{\gamma R}$$

$$T_t = T \left(1 + \frac{V^2}{2} \frac{\gamma-1}{\gamma R T} \right)$$

$$T_t = T \left(1 + \frac{V^2}{2} \frac{\gamma-1}{a^2} \right)$$

$$T_t = T \left(1 + M^2 \frac{\gamma-1}{2} \right)$$

Stagnation Properties

- Thermodynamic properties of a gas brought to rest isentropically and adiabatically

- Stagnation Pressure

$$p_t = p \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$$

- Stagnation Density

$$\rho_t = \rho \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{1}{\gamma-1}}$$

derive isentropic
flow equations

$$\frac{p_0}{p} = \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma-1}}$$

- What happens to these quantities in an isentropic process?

$$dh_r = 0$$

For no work or heat, all remain constant

Stagnation Properties

- Recall we said for the First Law (open system)
 - $d\left(h + \frac{1}{2}V^2\right) = \delta q - \delta w$
- Using the stagnation concept, we instead write
 - $dh_t = \delta q - \delta w$
- We no longer need to worry which term the heat and work modify (worry about it later)